

Neil Artus

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PROFESSIONAL SUMMARY

Software Engineer specializing in backend systems, Agentic AI architectures, and cloud deployments. Experienced in building production microservices, full-stack web applications, and automated CI/CD pipelines across Azure and bare-metal environments. Proven background in delivering high-impact automation tools, secure API integrations, and low-latency data pipelines.

SKILLS

Languages: Python, Go, C#, JavaScript, TypeScript, SQL

Web and Backend: FastAPI, Flask, React, Next.js, Vite, Tailwind CSS, .NET, Fiber, OpenTelemetry

Data and Databases: Cosmos DB, PostgreSQL, Qdrant, Microsoft SQL Server, Entity Framework, SQLite, Pandas, Pydantic

AI & Machine Learning: Azure OpenAI, Azure AI Foundry, LangChain, LangGraph, Semantic Kernel, Google Vertex AI, PyTorch, TensorFlow, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP)

Cloud & DevOps: Azure DevOps, Docker, Firebase, Hetzner, Nginx, Gunicorn, Ubuntu Linux, Windows Server 2022, CI/CD

Cybersecurity & Tools: Microsoft Entra ID (Azure AD), Okta, Role-Based Access Control (RBAC), SonarQube, Black Duck, Coverity, Cloudflare, Git

EXPERIENCE

Emerson | AI Engineer Intern | Metro Manila, Philippines | Mar 2026 – Jul 2026

- Automated cybersecurity compliance workflows using Azure AI Foundry Agents, custom hardened Model Context Protocol (MCP) integrations (Black Duck, OpenSSF, GitHub), and multi-agent orchestration graphs, saving **1,600+** analyst hours and **\$85,000+** in operational expenses.
- Built a real-time telemetry monitoring dashboard for AI agents and supply chain security metrics with sub-100ms p99 latency, establishing organizational secure development guidelines for MCP implementations.
- Upgraded and deployed security hotfixes for Emerson DeltaV compliance software using .NET 8, accelerating vulnerability mitigation and report generation speeds by 30%.
- Designed and maintained automated Azure DevOps CI/CD pipelines, reducing manual build and deployment cycles by 85%.
- Refactored legacy design system UI components using React, Radix UI, TanStack Table, and Recharts.
- Implemented enterprise authentication and access management using Okta, Microsoft Entra ID (Azure AD), and .NET middleware.

Albawani Co. | Freelance Software Engineer | Remote | Aug 2025 – Jan 2026

- Developed an internal HR automation application in Python that processes and generates attendance summary reports for **10,000+** employees, cutting manual reporting time by **50%** per pay cycle.
- Deployed and configured backend services on Hetzner bare-metal Linux servers, integrating directly with existing enterprise IT infrastructure.
- Implemented secure authentication and Role-Based Access Control (RBAC) using Microsoft Entra ID to safeguard employee personal records.

JG Summit Holdings Inc. | AI Engineer Intern (GenAI Research) | Metro Manila, Philippines | Jun 2025 – Aug 2025

- Developed and deployed a full-stack Business Intelligence dashboard using Azure OpenAI and LangChain, delivering automated financial reporting workflows to Treasury and Corporate Strategy teams.
- Built an Agentic RAG (Retrieval-Augmented Generation) pipeline across 12+ enterprise data sources, decreasing manual financial analysis turnaround by **70%**.
- Engineered automated data validation pipelines using SQL and Pydantic, processing 500+ monthly data points with **93%** validation accuracy.
- Authored modular system architecture blueprints and technical documentation, enabling handoff and production deployment by 1 senior and 2 junior engineers within 3 days.

PROJECTS

AGOS: Agentic Philippine Market Intelligence Platform | Full-Stack Developer

- Architected a multi-service market intelligence platform (React, FastAPI, Cosmos DB), engineering data ingestion pipelines that processed 350+ data points every 15 minutes for real-time stock forecasting, reducing ingestion latency by **over 60%**.
- Hardened microservices** using FastAPI dependency injection for server-side Firebase JWT verification, parameterized Cosmos DB queries to prevent injection attacks, and route-specific rate limiting via SlowAPI.
- Engineered a streaming financial research agent using LangGraph and Gemini API, managing 5 custom tool registries with sub-second response streaming.

MediSeen | Full-Stack Developer

- Led development of an Agentic AI progressive web application for clinical decision support, winning **1st Place** at the International Conference on Artificial Intelligence (ICAI 2026).
- Engineered cloud microservices on Azure, achieving 98 ms average database query latency and 400 ms time-to-first-token AI inference latency.
- Built a mobile-responsive interface using Tailwind CSS, improving measured System Usability Scale (SUS) scores by 15%.
- Configured security middleware protections, API routing policies, and Cloudflare DNS caching and SSL/TLS termination.

CNN-Osteo | Embedded AI Engineer

- Coordinated cross-functional development of a Convolutional Neural Network (CNN) using MobileNet and TensorFlow for automated osteoarthritis detection, earning **1st Place** at the Cavite Innovex 2025 Colloquium (COESCA Research of the Year).
- Trained and validated the model across 2,236 medical images, achieving **99.40%** validation accuracy and **98.19%** test accuracy.
- Deployed inference scripts to a Raspberry Pi 5 interfaced with a thermal camera module for real-time thermal diagnostic imaging.

Alersense | Full-Stack Engineer

- Built an IoT attention-monitoring platform using an **ESP32** microcontroller, **Python** data ingestion engine, **Flask-SocketIO**, and a **Next.js** real-time alert interface.
- Implemented real-time telemetry streaming for vital signs (heart rate, SpO2, body temperature), broadcasting low-latency alerts without page reloads.

Deterministic Finite Automata (DFA) Compiler | Full-Stack Developer

- Developed an interactive web-based DFA simulator and compiler using TypeScript, Flask, and SVG rendering to model state transitions and formal grammar logic in real time.
- Decoupled automata simulation algorithms from API routing logic, implementing test suites with pytest to verify strict compliance with formal language specifications.

EDUCATION

De La Salle University - Dasmariñas | Dasmariñas, Cavite, Philippines

Bachelor of Science in Computer Science with Specialization in Intelligent Systems (AI)

- **Academic Standing:** Magna Cum Laude | **GPA:** 3.52 / 4.0 | **TOEIC:** 955 / 990
- **Leadership & Activities:** Secretary & Codeyssey Program Head – Computer Science Program Council (CSPC); managed technical workshops and hackathons for 1,000+ students
- **Honors:** CSPC Leadership Award Recipient, Dean's Honor List
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Database Systems, Web Development, Computer Networks, Operating Systems, Software Engineering